Optimal Gaussian Approximation for Stationary Spatial Fields

Soham Bonnerjee
University of Chicago

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This is joint work with



Sayar Karmakar
University of Florida

1 Problem Introduction

2 Main Results

3 References

Central Limit Theory and more

$(X_{\boldsymbol{i}} : \boldsymbol{i} = (i_1, i_2, \cdots, i_d) \in \mathbb{Z}^d)_{\mathbf{1} \leq \boldsymbol{i} \leq \boldsymbol{n}}$ are i.i.d random fields with finite variance $\sigma^2 = \mathbb{E}[X_{\boldsymbol{i}}^2] < \infty$. Let $|\boldsymbol{i}| = i_1 \cdots i_d$. Central Limit Theorem

$$\sqrt{|\boldsymbol{n}|}\bar{X} - N(0, \sigma^2) \xrightarrow{p} 0 \iff \frac{|S_{\boldsymbol{n}} - N(0, \mathbb{E}[S_{\boldsymbol{n}}^2])|}{\sqrt{|\boldsymbol{n}|}} \xrightarrow{p} 0.$$

For simplicity assume $n_1 = n_2 = \cdots = n_d = n$, so that $|\boldsymbol{n}| = n^d$.

Central Limit Theory and more

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- Functional CLT for i.i.d. random field[Wichura(1969)]

$$\max_{\mathbf{1} \le \boldsymbol{i} \le \boldsymbol{n}} \frac{|S_{\boldsymbol{i}} - \mathbb{B}(\mathbb{E}[S_{\boldsymbol{i}}^2])|}{n^{d/2}} \xrightarrow{p} 0.$$

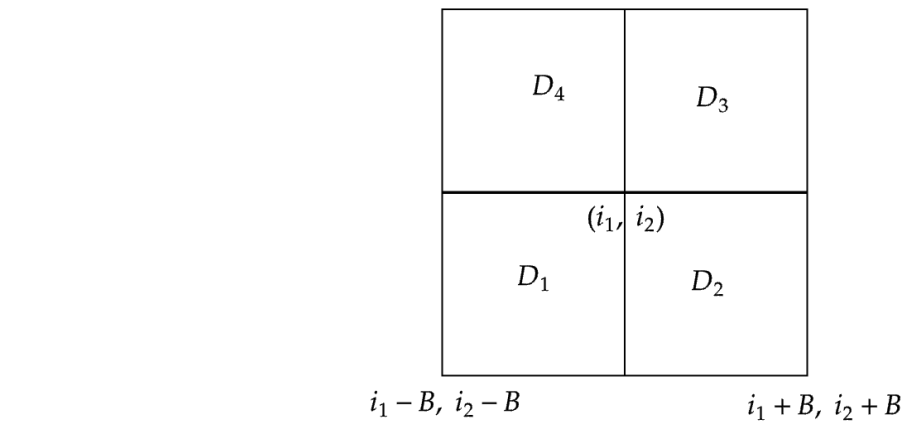
- Stationary **associated** (satisfies FKG inequalities): Newman(1980), Bulinski and Keane(1996), Balan(2005) .
  Mixing: Berkes and Morrow(1981), Bolthausen(1982) .
- No KMT-type result + No result for stationary causal processes.

Why is it important? Spatial Change-Point Detection Problem

- For $d = 1$, change-point in time series: very well studied.

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- For $d = 1$, change-point in time series: very well studied.
- For even $d = 2$, it is complicated.



Two-dimensioanl MOSUM : $T_{\boldsymbol{i}} = \sum_{cyc}(D_1 - D_2)^2$, and test statistic: $G = \max_{\mathbf{1} \le \boldsymbol{i} \le \boldsymbol{n}} T_{\boldsymbol{i}}$. Han, 2021.

Why is it important?

Suppose

$$\max_{\mathbf{1} \leq \boldsymbol{i} \leq \boldsymbol{n}} \frac{|S_{\boldsymbol{i}} - \sigma \mathbb{B}_{\boldsymbol{i}}|}{\tau_n} \xrightarrow{p} 0 \tag{1}$$

holds for some rate τ_n.

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- Let $W := \sup_{t\in(0,1)} W(t) = \sup_{t\in(0,1)} \sum_{\mathbf{1}\leq \boldsymbol{i}\leq \boldsymbol{n}} w_i(t) X_i$. G is a function of W.

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- Estimate $W(t)$ by $W^{\diamond}(t) := \sum_{\mathbf{1}\leq \boldsymbol{i}\leq \boldsymbol{n}} w_i(t)\sigma(\mathbb{B}_{\boldsymbol{i}} - \mathbb{B}_{\boldsymbol{i}-(1,0)} - \mathbb{B}_{\boldsymbol{i}-(0,1)} + \mathbb{B}_{\boldsymbol{i}-(1,1)})$.

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- Use (1) to get rate on $\sup_{t\in(0,1)} |W(t) - W^{\diamond}(t)|$.
- Simulating $W^{\diamond}$ is very easy using bootstrap!!

Main theorem set-up: A general class of time-series

Suppose random variables. Non-linear dependence

- Consider stationary AR(1)

$$X_i = aX_{i-1} + e_i.$$

- This can be rewritten as

$$X_i = \sum_{j=0}^{\infty} a^j e_{i-j}.$$

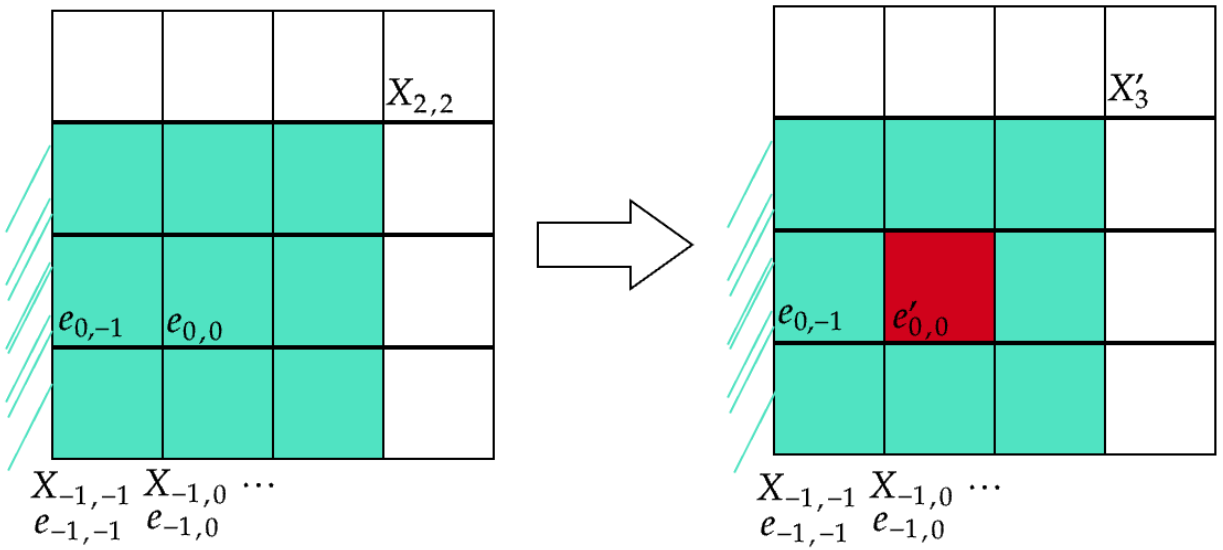
- General Nonlinear stationary causal process:

$$X_i = H(e_i, e_{i-1}, \ldots)$$

- Non-stationary:

$$X_i = H_i(e_i, e_{i-1}, \ldots)$$

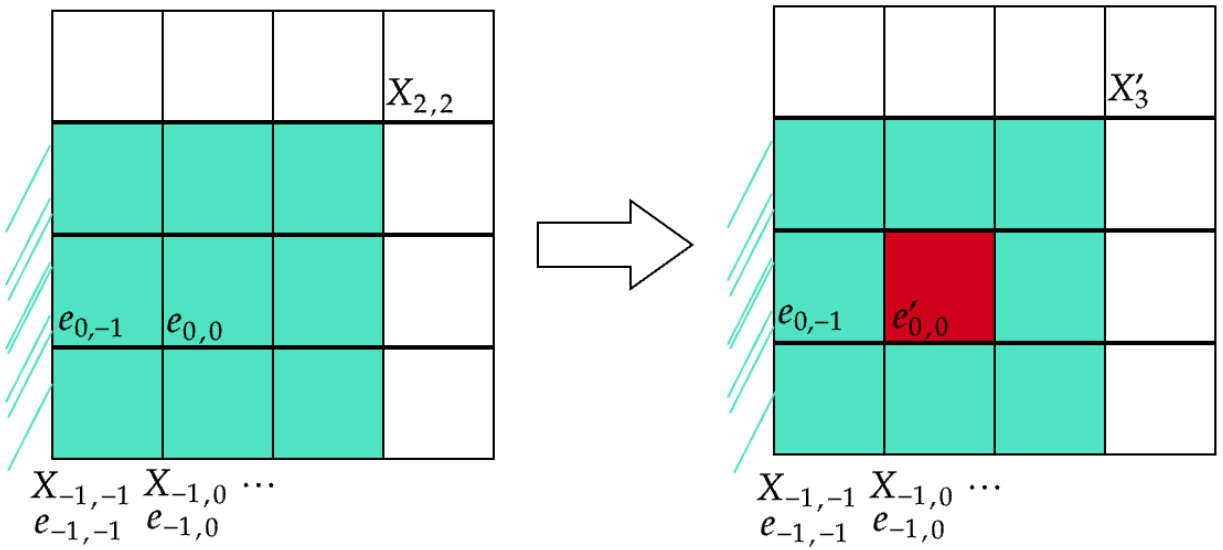
Main theorem set-up: In search of coefficients



$X_i = g(e_s : s \leq i)$

Flip $e_{0,0}$ to an i.i.d. copy $e'_{0,0}$
Get new X'_i

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Get new X'_i

- **Functional dependence measure** [Machkouri et. al. 2013, Deb et. al. 2017]:-

$$\delta_{\boldsymbol{i},p} = \|X_{\boldsymbol{i}} - X_{\boldsymbol{i}'}\|_p \ , \ \boldsymbol{i} \in^d . \tag{2}$$

- Thus equivalently we can now impose conditions on how $\theta_{j,p}$ decays. We put $\Theta_{m,p} = \sum_{\|\boldsymbol{i}\|_\infty \geq m} \delta_{\boldsymbol{i},p} = O(m^{-A})$.

Main Theorem

Result we want to prove: $\mathbb{B}$: Brownian Sheet:

- For fixed $t \in^d$, $\mathbb{B}_{\boldsymbol{t}} \sim N(0, |\boldsymbol{t}|)$, and
- $\mathrm{Cov}(\mathbb{B}_{\boldsymbol{s}}, \mathbb{B}_{\boldsymbol{t}}) = \prod_{i=1}^{d}(s_i \wedge t_i)$.

Theorem (Informal)

If $\sigma^2 := \sum_{\boldsymbol{k} \geq \mathbf{0}} \mathbb{E}[X_{\mathbf{0}} X_{\boldsymbol{k}}] > 0$ *then under fast enough decay i.e* $A > A_0(p)$,

$$\max_{\mathbf{1} \leq \boldsymbol{i} \leq \boldsymbol{n}} |S_{\boldsymbol{i}} - \sigma \mathbb{B}_{\boldsymbol{i}}| = \mathrm{o}_{\mathbb{P}}(n^{d/p}).$$

For $d = 1$, see KMT(1975), Shakhanenko(2006), Berkes, Liu and Wu (2014), Karmakar and Wu(2021) , Bonnerjee, Karmakar and Wu (2023).

Truncation step

Truncation: to control the stationary sequences well:

$$X_{\boldsymbol{i}}^{\oplus} = T_{n^{d/p}}(X_{\boldsymbol{i}}), \text{ where } T_b(w) = \max\{\min\{w, b\}, -b\} \tag{3}$$

Truncated partial sum process $\{S_i^{\oplus}\}_{i=1}^n$:

$$S_{\boldsymbol{i}}^{\oplus} := \sum_{\mathbf{1} \le \boldsymbol{j} \le \boldsymbol{i}} \left(X_{\boldsymbol{i}}^{\oplus} - \mathbb{E}(X_{\boldsymbol{i}}^{\oplus}) \right). \tag{4}$$

Then, $\max_{\mathbf{1} \le \boldsymbol{i} \le \boldsymbol{n}} |S_{\boldsymbol{i}} - S_{\boldsymbol{i}}^{\oplus}| = o_{\mathbb{P}}(n^{d/p})$.

m-dependence

$$\tilde{X}_{\boldsymbol{i}} = \mathbb{E}(X_{\boldsymbol{i}}^{\oplus} \mid \varepsilon_{\boldsymbol{i}-\boldsymbol{s}} : |\boldsymbol{s}|_{\infty} \leq m) - \mathbb{E}(X_{\boldsymbol{i}}^{\oplus}). \tag{5}$$

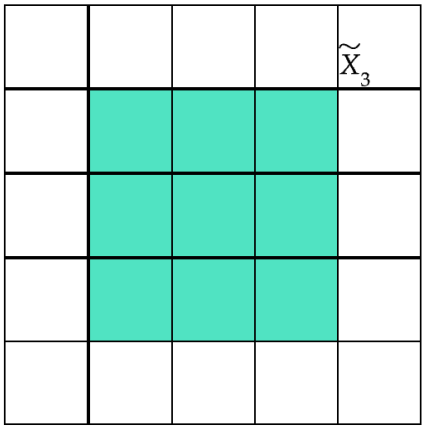
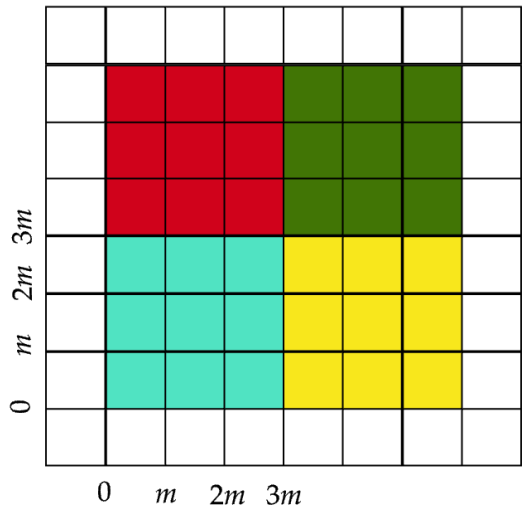


Figure: m-dependence with $m = 3$

It can be proven:

$$\| \max_{\mathbf{1} \leq \boldsymbol{i} \leq \boldsymbol{n}} |S_{\boldsymbol{i}}^{\oplus} - \tilde{S}_{\boldsymbol{i}}| \|_p \leq c_p n^{d/2} \Theta_{1+m,p} = o(n^{d/p}). \tag{6}$$

Blocking Step

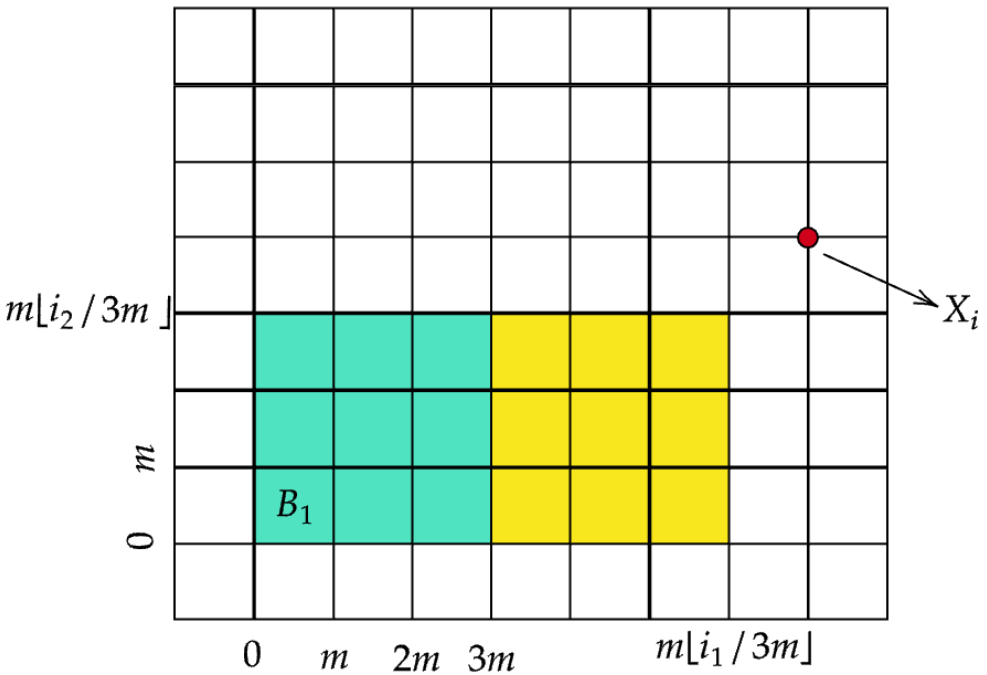


Small Blocks of size $m \rightarrow$ Big blocks of size $3m$.

Blocking Approximation

$\boldsymbol{l} \geq \mathbf{0}$, let $B_{\boldsymbol{l}} := \sum_{\boldsymbol{i}:i_k=(l_k-1)m+1}^{l_k m} \tilde{X}_{\boldsymbol{i}}$. Let $\boldsymbol{l_i} = \lfloor \boldsymbol{i}/3m \rfloor$, and

$$S_{\boldsymbol{i}}^{\diamond} = \sum_{\mathbf{1} \leq \boldsymbol{l} \leq \boldsymbol{l_i}} \sum_{3\boldsymbol{l}-2 \leq \boldsymbol{k} \leq 3\boldsymbol{l}} B_{\boldsymbol{k}}. \tag{7}$$

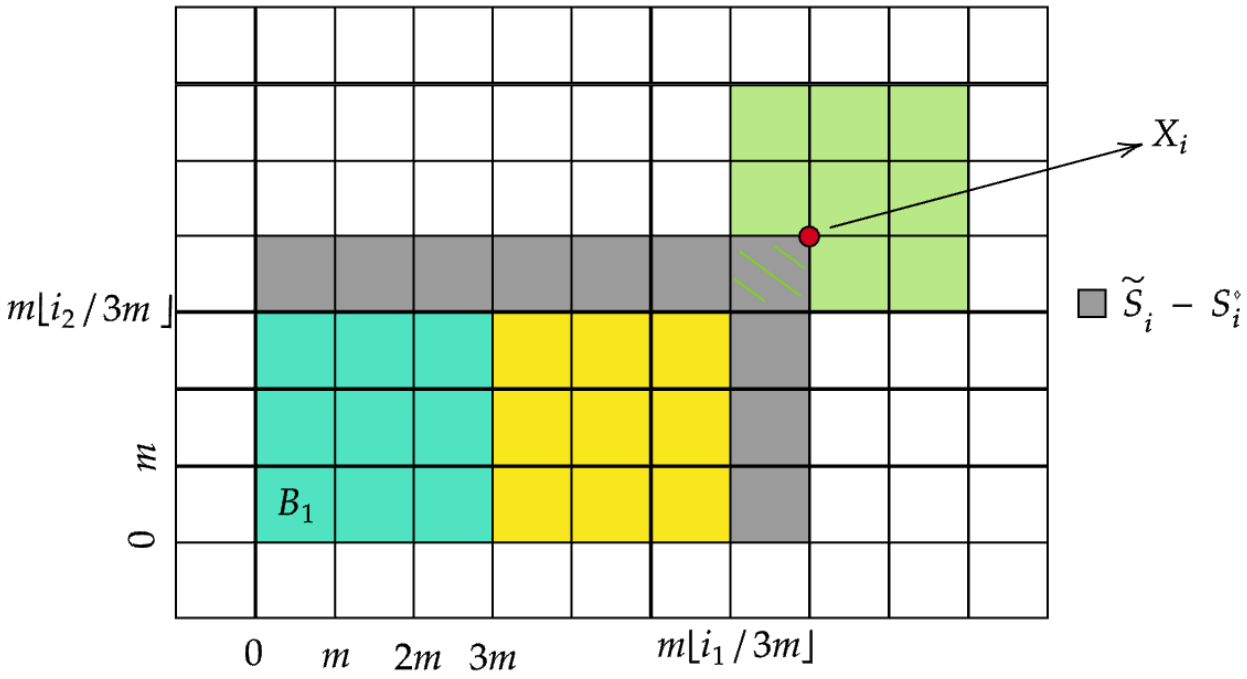


Blocking Approximation (Ctd)

Need to prove: $\max_{\mathbf{1} \le \boldsymbol{i} \le \boldsymbol{n}} |\tilde{S}_{\boldsymbol{i}} - S^{\diamond}_{\boldsymbol{i}}| = o_{\mathbb{P}}(n^{d/p})$.

Blocking Approximation (Ctd)

Need to prove: $\max_{\mathbf{1}\le \boldsymbol{i}\le \boldsymbol{n}} |\tilde{S}_{\boldsymbol{i}} - S^{\diamond}_{\boldsymbol{i}}| = o_{\mathbb{P}}(n^{d/p})$. What is $|\tilde{S}_{\boldsymbol{i}} - S^{\diamond}_{\boldsymbol{i}}|$?



All the points in the green shaded area has the blue and yellow shaded area as blocking approximation.

Let $R_{\boldsymbol{i}}$: grey shaded area.

Blocking Approximation (Ctd)

Theorem

For $\boldsymbol{j} \geq \boldsymbol{k} \geq \mathbf{0}$, *define* $R_{\boldsymbol{i}} = \sum_{\boldsymbol{k} \leq \boldsymbol{l} \leq \boldsymbol{i}} X_{\boldsymbol{l}}$ *for* $\boldsymbol{k} \leq \boldsymbol{i} \leq \boldsymbol{j}$. *Let* $n := |\boldsymbol{j} - \boldsymbol{k}|$ *and* $R^\star := \max_{\boldsymbol{k} \leq \boldsymbol{i} \leq \boldsymbol{j}} R_{\boldsymbol{i}}$. *Then,*

$$\|R^\star\|_p \lesssim n^{1/2}\left(\sum_{\boldsymbol{s} \in G_{\boldsymbol{j}-\boldsymbol{k}}} \delta_{\boldsymbol{s},2} + \sum_{\boldsymbol{s} \in G^c_{\boldsymbol{j}-\boldsymbol{k}}} \delta_{\boldsymbol{s},p} + \|X_{\boldsymbol{i}}\|_2\right) + n^{1/p}\left(\sum_{\boldsymbol{s} \in G_{\boldsymbol{j}-\boldsymbol{k}}} \|\boldsymbol{s}\|_\infty^{d/2-d/p} \delta_{\boldsymbol{s},p} + \|X_{\boldsymbol{i}}\|_p\right),$$

where $G_{\boldsymbol{l}} = \{\mathbf{1} \leq \boldsymbol{i} \leq \boldsymbol{l}\}$.

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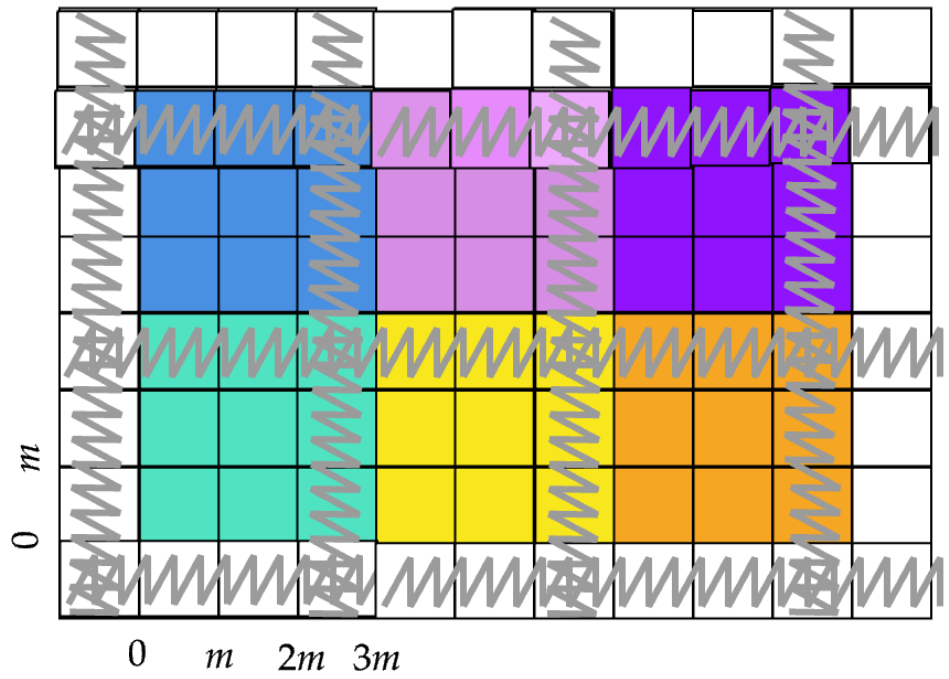
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where $G_{\boldsymbol{l}} = \{\mathbf{1} \leq \boldsymbol{i} \leq \boldsymbol{l}\}$.

This helps us in getting $\max_{\mathbf{1} \leq \boldsymbol{i} \leq \boldsymbol{n}} |\tilde{S}_{\boldsymbol{i}} - S^{\diamond}_{\boldsymbol{i}}| = o_{\mathbb{P}}(n^{d/p})$.

Conditional Gaussian approximation

Now we have blocks of size $3m$. Let blocking approximation of $S_{\boldsymbol{i}}$ be $\mathcal{B}_{\boldsymbol{i}/3m}$.



Conditional on shaded boundary regions, the blocks are independent. So we need a gaussian approximation result for independent random field.

Required Result

We need a result like: Let X_i be independent random field with finite variance. Then

$$\max_{\mathbf{1} \leq \boldsymbol{i} \leq \boldsymbol{n}} |S_{\boldsymbol{i}} - \mathbb{B}(\mathbb{E}[S_{\boldsymbol{i}}^2])| = o_{\mathbb{P}}(n^{d/p}). \tag{8}$$

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If we have such a result, we can use it twice: once on the conditional block, and then after lifting the condition. We can prove that resulting brownian sheet from these two successive Gaussian approximation, is close to $\sigma\mathbb{B}_{\boldsymbol{i}}$. This will prove the theorem.

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THANK YOU !!

QUESTIONS?